FALCATA SYSTEMS

UAS & Geospatial Intelligence - Built for the field, not the brochure.

WHO WE'RE BUILDING FOR

Portuguese Army | Portuguese Air Force | Civil Protection

We're designing Amélia Nótus around the real operational demands of tactical reconnaissance, airborne systems integration, and emergency terrain survey - the environments our target users actually operate in.

CURRENT STAGE

We're currently building our first Amélia Nótus prototype - sourcing core hardware, including LiDAR, radar, and flight control components, now, with a first field test targeted for this December. Our focus is hostile-environment performance and mission-critical reliability from day one.

TECHNOLOGY FOCUS

GPS-DENIED NAVIGATION | LIDAR INTEGRATION | RESILIENT COMMS | AUTONOMOUS FLIGHT | GEOAI

PLATFORM CAPABILITIES

- GPS-denied navigation: onboard visual odometry, fused with inertial and barometric sensing, corrects drift and maintains accurate position and speed without a satellite link; reference-map matching and route relocalisation extend this into a full GNSS-denied fix.
- Clean value chain: electronics sourced exclusively from allied nations (NATO, Taiwan, Japan), avoiding espionage risk and vetoed hardware.
- Modular sensor payloads: interchangeable sensor bay supporting multiple ranging technologies for topography and obstacle detection.
- Lean manufacturing: in-house PCB prototyping and 3D-printed aero-grade chassis, cutting unit costs well below traditional defense-contractor pricing.

FUTURE PIPELINE

Geospatial Intelligence Platform

Drone-based airborne surveys - magnetometry, LiDAR, gamma-ray spectrometry - processed with machine learning for terrain analysis and threat assessment.

Managed Survey Services

Full-cycle airborne data collection and reporting for defence and civil protection organizations, from flight planning through delivered analysis.

COMPLIANCE & TRUST POSTURE

Regulatory & Certification: we are not yet certified under ANAC/EASA drone regulations. Pursuing the appropriate certification pathway is part of our roadmap as we move toward operational testing.

Export Control & Procurement: we have not yet established formal export-control or NATO STANAG compliance status. We intend to build this posture in step with our first operational partnerships.

FOUNDERS

Mike - Founder, Sensor Integration & Software Architecture

Convinced a vision sensor and a LiDAR module to stop arguing and cooperate - basically diplomacy. When the platform doesn't need GPS to know where it is, that's Mike.

Nuno - Founder, Hardware, PCB & Manufacturing

Builds the part that has to survive being dropped, shaken, rained on, and occasionally yelled at - chassis, CAD models, and PCB design included.

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